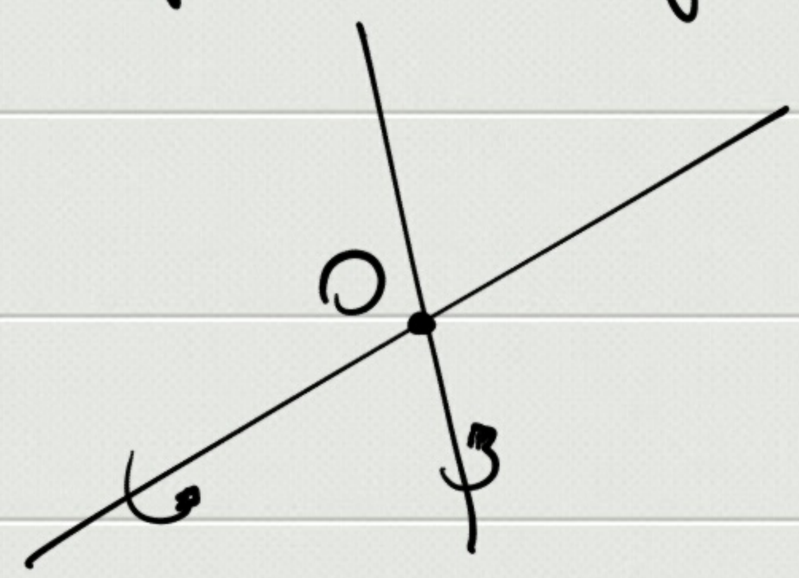
Giroscopio : c.r. un punto è fisso

$\Rightarrow$ moto rotatorio

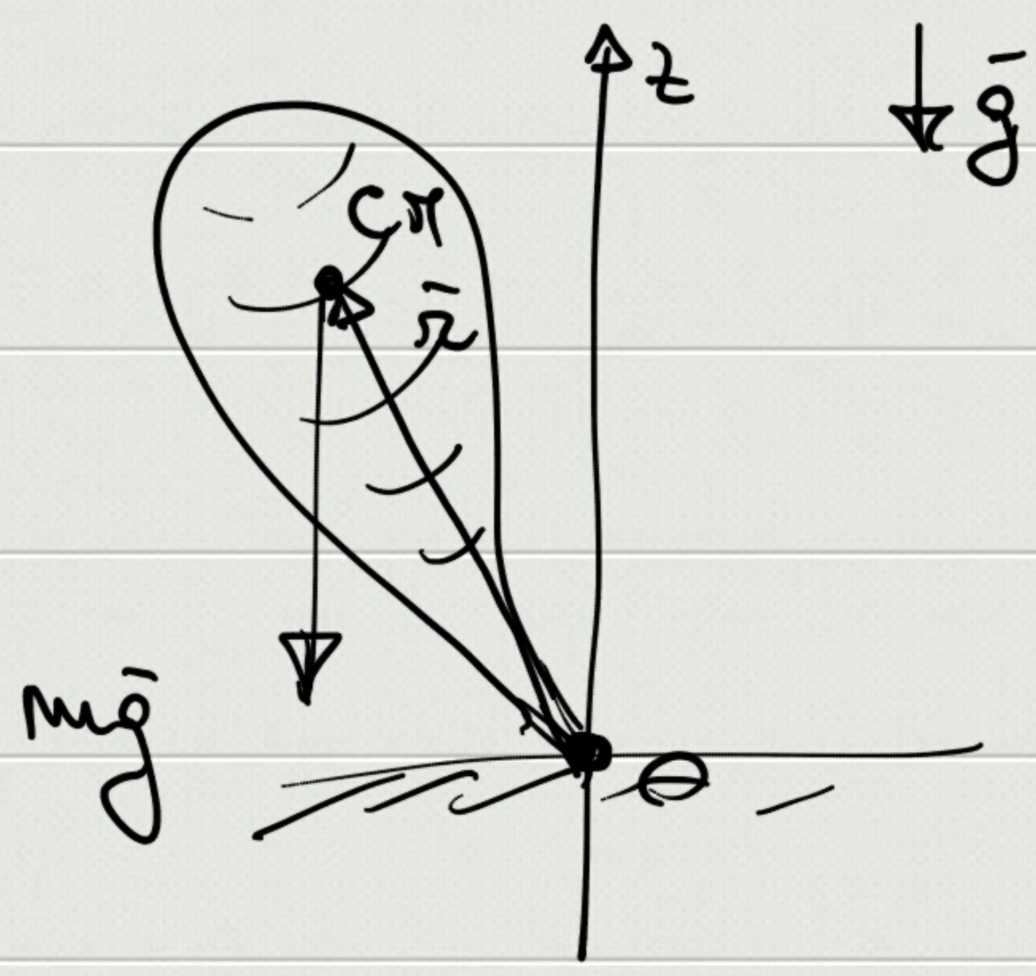


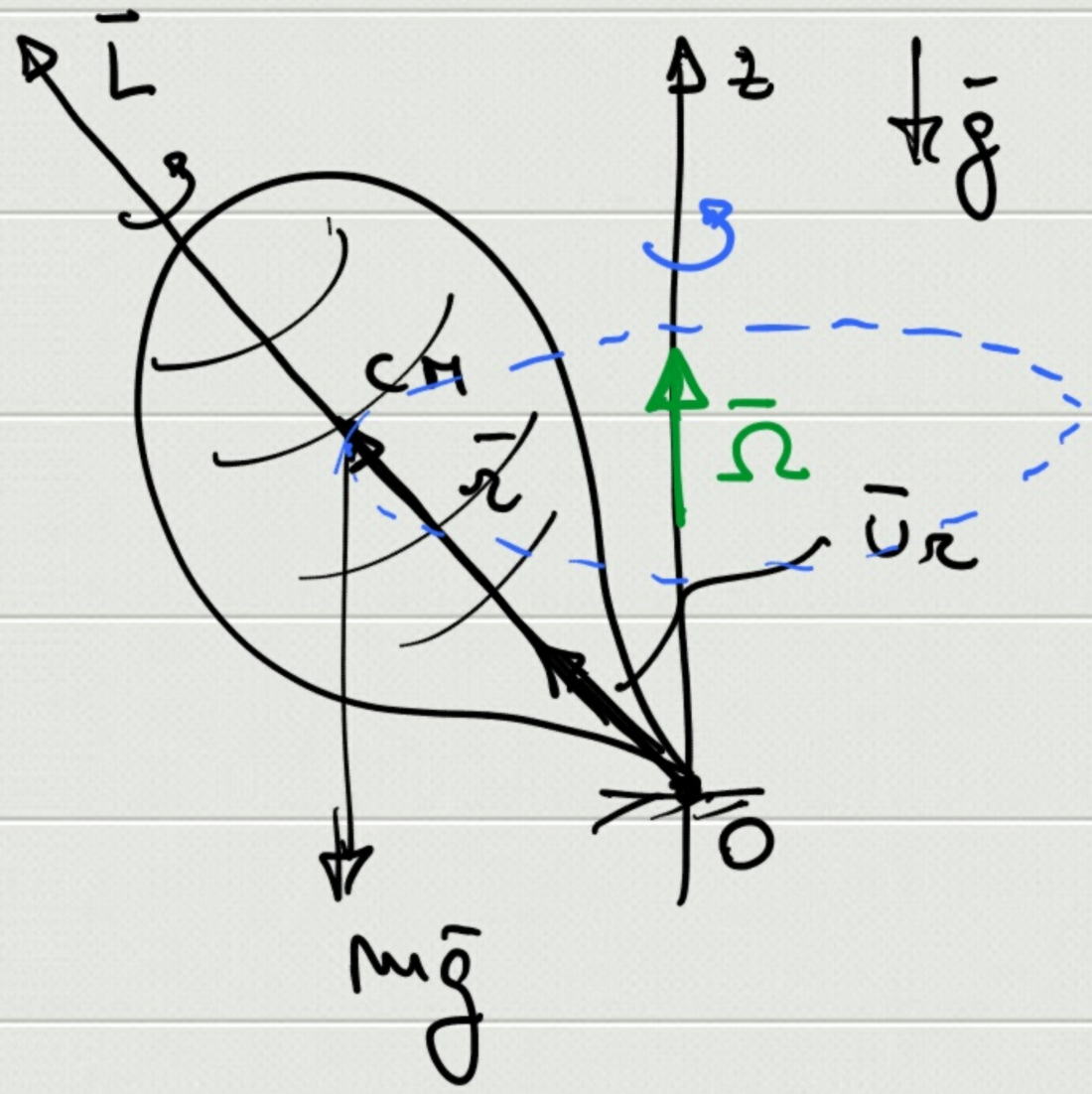
- $O \equiv CM$ $\quad \bar{M}_O^E = 0$ $\quad$ asse principale di inerzia

$\Rightarrow \bar{L}_O = I\bar{\omega}$

$$\bar{M}_O^E = \frac{d\bar{L}_O}{dt} = 0 \Rightarrow \boxed{\bar{L}_O = \text{cost}}$$

- $O \neq CM$ $\quad \bar{M}_{O,peso} \neq 0$





$$\bar{\omega} = \omega \bar{u}_r$$

$$\bar{L}_o = \bar{I}\,\bar{\omega}$$

$$|\bar{\omega}| = \text{cost} \Rightarrow |\bar{L}_o| = \text{cost}$$

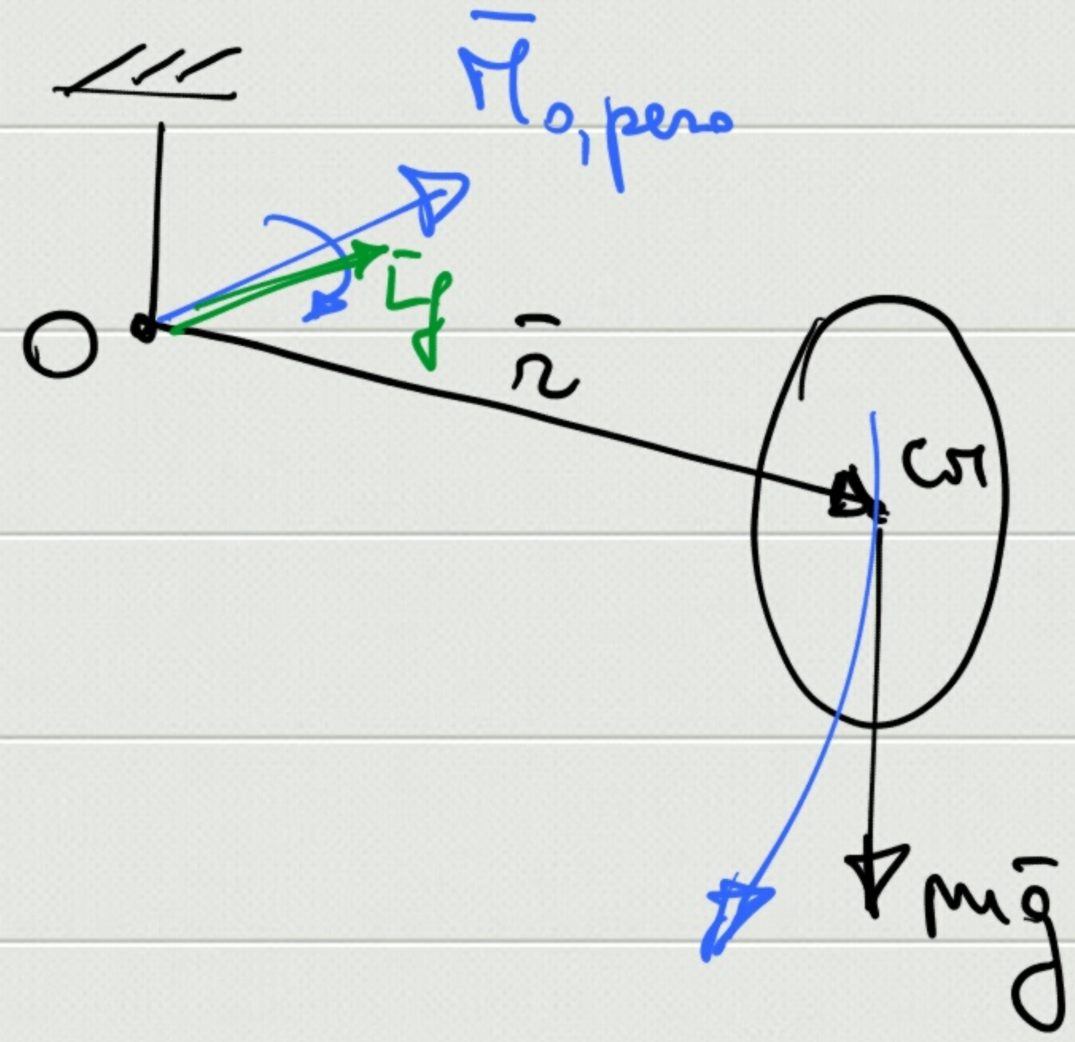
$$\frac{d\bar{L}_o}{dt} = \bar{\Omega} \times \bar{L}_o$$

$$\bar{M}_o^e = \frac{d\bar{L}_o}{dt} \Rightarrow \bar{r} \times m\bar{g} = \bar{\Omega} \times \bar{L}_o$$

$$-m\bar{g} \times \bar{r} = \bar{\Omega} \times \bar{I}\,\bar{\omega} \Rightarrow -m\bar{g} \times r\bar{u}_r = \bar{\Omega} \times \bar{I}\omega\bar{u}_r$$

$$\Rightarrow -mr\bar{g} \times \bar{u}_r = I\omega\bar{\Omega} \times \bar{u}_r$$

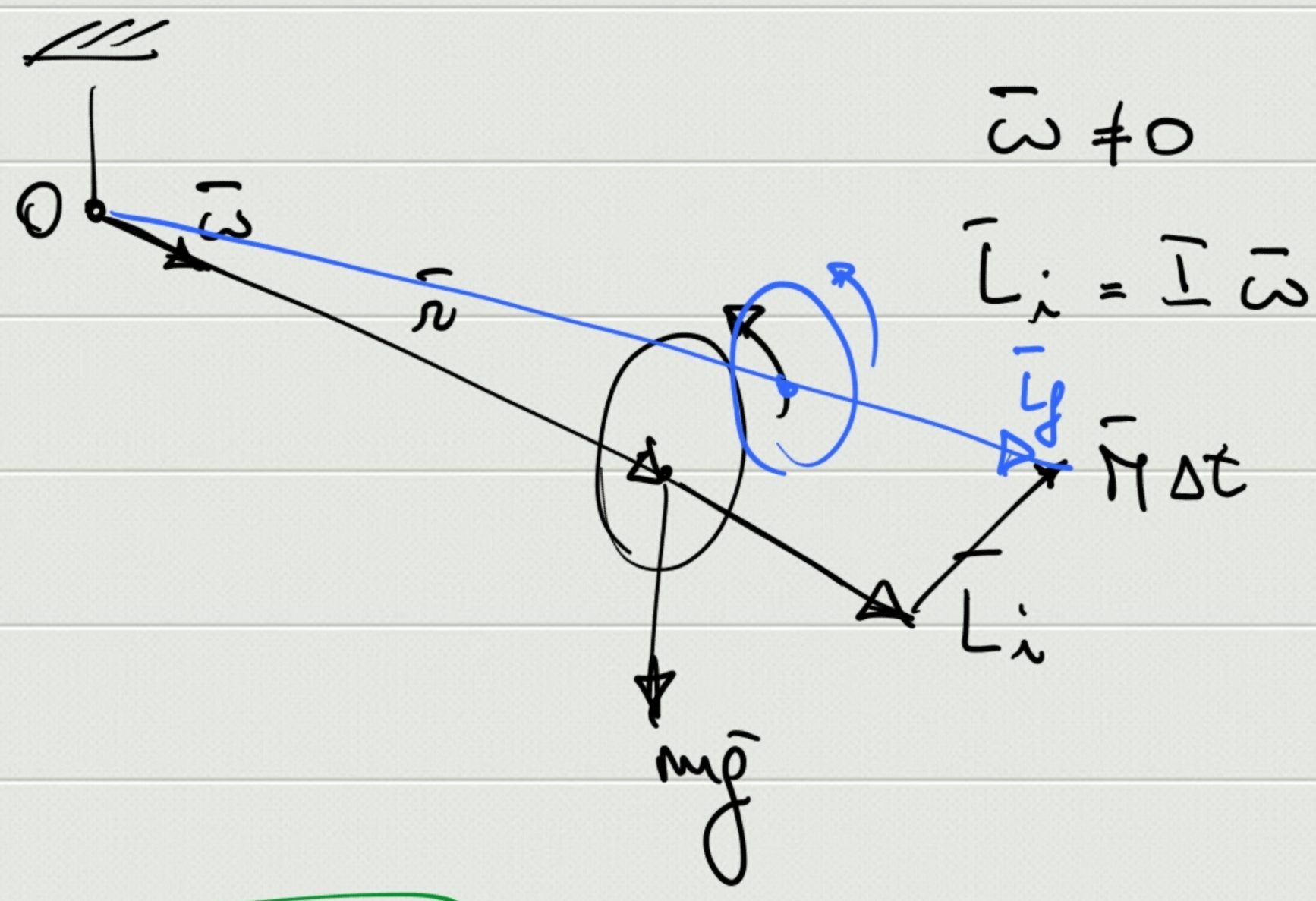
$$\Rightarrow -mr\bar{g} = I\omega\bar{\Omega} \Rightarrow \boxed{\bar{\Omega} = -\frac{mr}{I\omega}\bar{g}}$$



$$\bar{M}_{o,peso} = \bar{r} \times m\bar{g}$$

$$\int \bar{M}\, dt = \Delta \bar{L} = \bar{L}_f - \bar{L}_i$$

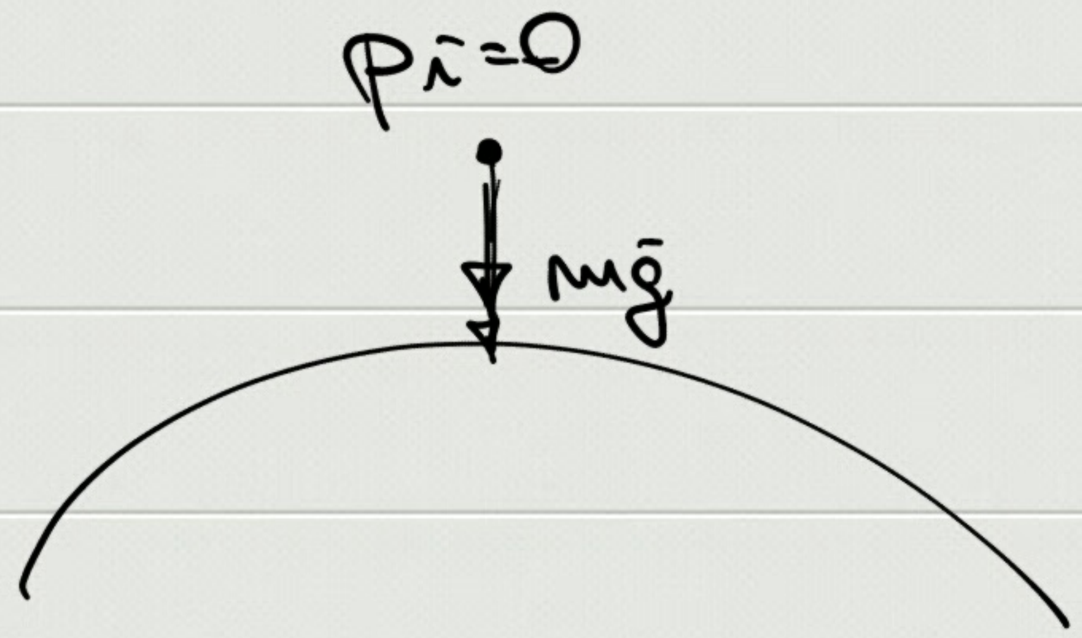
$$\Rightarrow \bar{L}_f = \underbrace{\bar{L}_i}_{=\,0} + \bar{M}\Delta t = \underbrace{\bar{r} \times m\bar{g}}_{\bar{M}_{o,peso}}\, \Delta t$$

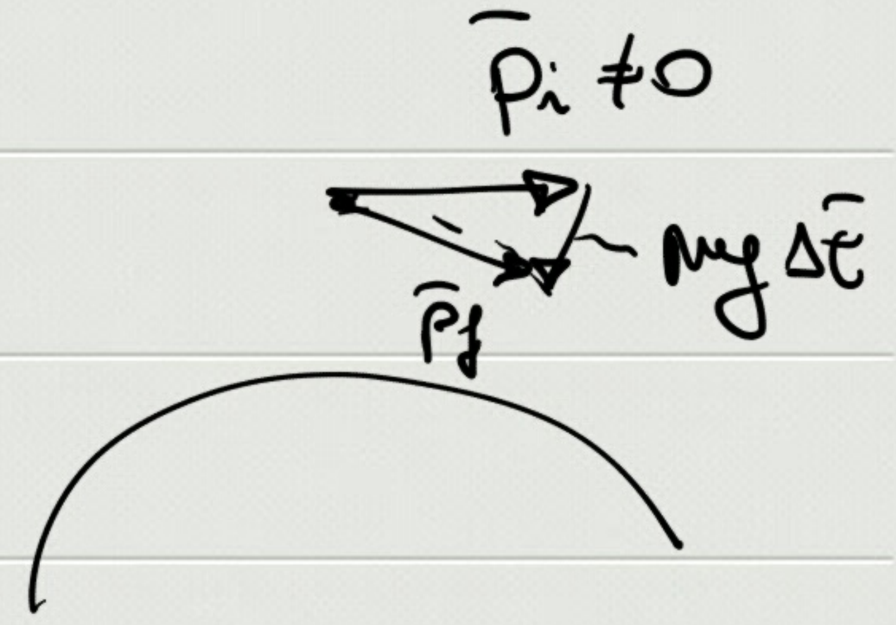


$$\bar{\omega} \neq 0$$

$$\bar{L}_i = \bar{I}\,\bar{\omega}$$

$$\boxed{\bar{L}_f = \bar{L}_i + \bar{M}\Delta t}$$





$$\bar{F}\,\Delta t = \Delta\bar{p} \implies \bar{p}_f = \bar{p}_i + m\bar{g}\,\Delta t$$